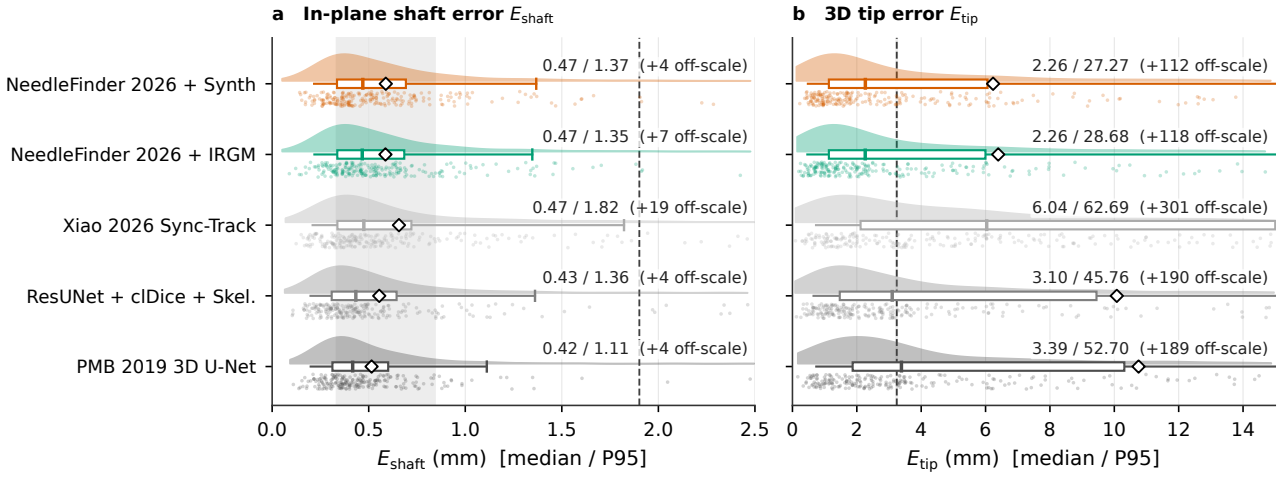

a In-plane shaft error E_{shaft}
b 3D tip error E_{tip}
NeedleFinder 2026 + Synth
NeedleFinder 2026 + IRGM
Xiao 2026 Sync-Track
ResUNet + clDice + Skel.
PMB 2019 3D U-Net
0.47 / 1.37 (+4 off-scale)
0.47 / 1.35 (+7 off-scale)
0.47 / 1.82 (+19 off-scale)
0.43 / 1.36 (+4 off-scale)
0.42 / 1.11 (+4 off-scale)
2.26 / 27.27 (+112 off-scale)
2.26 / 28.68 (+118 off-scale)
6.04 / 62.69 (+301 off-scale)
3.10 / 45.76 (+190 off-scale)
3.39 / 52.70 (+189 off-scale)
0.0
0.5
1.0
1.5
2.0
2.5
0
2
4
6
8
10
12
14
E_{shaft} (mm) [median / P95]
E_{tip} (mm) [median / P95]